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DEPARTMENT OF DEFENSE HANDBOOK

ELECTROMAGNETIC ENVIRONMENTAL EFFECTS TO ORDNANCE TEST GUIDE

PART 7

HAZARDS OF ELECTROMAGNETIC RADIATION TO ORDNANCE OPERATIONAL GUIDANCE

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FOREWORD

1. This handbook is approved for use by all Departments or Agencies of the Department of Defense (DoD).

2. This part supplements MIL-HDBK-240-1 and was prepared for contractors, DoD test agencies, activities, and test planners. This handbook is for guidance only and cannot be cited as a requirement.

3. This handbook provides recommended practices for conducting ordnance operations in the presence of the electromagnetic environment (EME). Developed over many years by DoD test activities, the methods and procedures described herein are the product of the collective DoD Electromagnetic Environmental Effects (E3) test experience. This handbook consolidates these practices into a single document vice numerous individual Service standards, instructions, thus promoting a standardized approach.

4. This handbook was prepared by the Services under the sponsorship of the Joint Ordnance Commander's Group (JOCG) Electromagnetic Environmental Effects (E3) Ordnance Safety Subgroup in accordance with the guidelines of the Standardization Reform Policy established by the Secretary of Defense.

5. Comments, recommendations, additions, or deletions and any other pertinent data that may improve this document should be emailed to marcus.shellman.civ@mail.mil or addressed to:

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1. SCOPE

1.1 Purpose. This part of MIL-HDBK-240 provides the recommended methods and techniques to develop the guidance necessary to establish electromagnetic environment (EME) controls to reduce the risk of hazards of electromagnetic radiation to ordnance (HERO). This part also supplements Defense Explosives Safety Regulation (DESR) 6055.09 by providing the guidance necessary to establish EME controls to ensure HERO is addressed during military operations or training exercises.

1.2 Background. There are six specific objectives of this handbook:

a. Document electromagnetic environmental effects (E3) Joint-Service methodology.

b. Promote standardization.

c. Provide operational guidance methodology for managing HERO.

d. Define HERO classifications and associated maximum allowable environments (MAEs) as defined in MIL-HDBK-240-2, Section 7.

e. Determine the safe separation distances (SSDs) for HERO SUSCEPTIBLE and HERO UNSAFE ORDNANCE.

f. Facilitate the exchange of data.

Ordnance containing electrically initiated devices (EIDs) are classified as either HERO SAFE, HERO SUSCEPTIBLE, or HERO UNSAFE. These general HERO classifications provide a means for identifying the level of susceptibility to the EME and are defined in MIL-STD-464.

In general, risk-reduction measures developed using analytical methods may entail silencing all emitters during ordnance operations, ensuring ordnance operations are conducted in radio frequency (RF)-free environments, or imposing SSDs between the offending emitters and HERO SUSCEPTIBLE and HERO UNSAFE ORDNANCE. These methods can be very restrictive and do not always allow for operational flexibility; the use of instrumented HERO surveys (measurements) to characterize the EME provides a better means for managing HERO (see MIL-HDBK-240-6). Ultimately, the information gathered is used to determine the operational guidance for the safe storage and use of ordnance and weapon systems containing EIDs.

When defining the operational EME, one should anticipate not only the individual Service EME, but also the EME in Joint-operation scenarios. The JCIDS Manual states that, “Because all weapons/weapon systems have the potential of being deployed together or employed in joint environments, weapons and weapon systems will be considered joint systems within the Joint Capabilities Integration and Development System (JCIDS) process and may be designated as JROC or JCB Interest.” As Joint operations become more commonplace, there is an increased likelihood that weapons from one Service will be exposed to other Services’ EME levels. MIL-STD-464 specifies the EME levels for HERO SAFE certification.

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2. APPLICABLE DOCUMENTS

2.1 General. See MIL-HDBK-240-1. This document can be found on ASSIST Online at https://assist.dla.mil.

3. DEFINITIONS

3.1 Definitions. See MIL-HDBK-240-1.

4. MANAGING HERO DURING ORDNANCE OPERATIONS

4.1 Introduction. The process for managing potential electromagnetic radiation (EMR) hazards and operational risk through proper emission control and/or ordnance handling procedures is covered in this part; specifically, maintaining HERO SSDs between ordnance and transmitting antennas, developing emission control procedures, and implementing general HERO requirements (for example, shielding/shunting ordnance electrical connectors, transporting ordnance in sealed, all-metal containers, and other risk mitigations). This part also discusses how to determine the HERO classification of ordnance as well as managing HERO during North Atlantic Treaty Organization (NATO) operations. The general HERO requirements apply to all ordnance items containing EIDs regardless of their HERO classification. The intent of this part is to provide specific guidance for operations involving HERO SUSCEPTIBLE or HERO UNSAFE ORDNANCE when such ordnance is exposed to EMEs.

Caution: Each facility or platform must be aware of all the emitters affecting ordnance. There are cases where the EME generated by some emitters exceed the HERO SAFE levels established by MIL-STD-464. In these cases, the operational guidance will need to provide SSDs or emission controls for HERO SAFE ordnance.

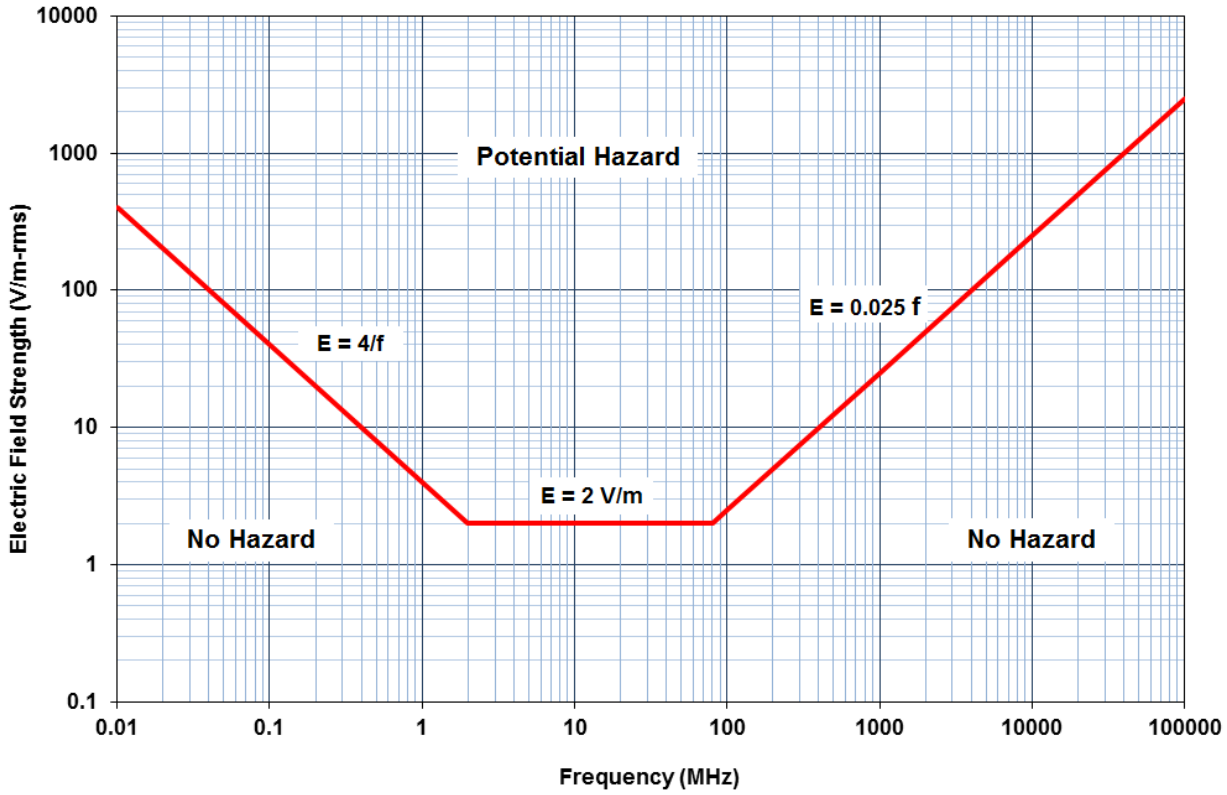
4.2 Management. Generally, managing operations involves using both the ordnance MAE (or HERO classification) and the measured and/or calculated EME from fixed (or permanently located) emitter systems. With these two critical pieces of information, HERO can be managed either through emission control procedures (for example, frequency and/or power management or sectoring) or through calculated SSDs. The latter option of calculating SSDs is purely analytical and is generally a more conservative approach for managing HERO (particularly in a shipboard environment) and the result is generally more restrictive and provides less operational flexibility. Historically, for the Navy, HERO surveys have provided the added benefit of measuring the actual EME that exists at ordnance locations and providing a more exact means for managing HERO on the deck of a ship to mitigate the HERO issue and provides the warfighter precise guidance for specific ordnance evolutions. That said, in the shipboard environment, SSDs are calculated to assist with the creation of the HERO emission control procedures that mitigate HERO for in-flight aircraft carrying ordnance; and to manage HERO issues for aircraft, mobile, and portable emitters that have no fixed location.

In order to determine the calculated SSD for an emitter, specific characteristics, such as average transmitter power, operational frequency range, and antenna gain must be identified. Then the MAE or the limits provided in the HERO curves for SUSCEPTIBLE and UNSAFE ORDNANCE (FIGURE 1 or FIGURE 2) must be identified. By manipulating the Friis free space equation and the above transmitter/antenna characteristics, the desired MAE data can be

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used to determine the distance at which the EME will exceed the susceptibility limit. HERO classifications, along with the EME levels in ordnance operating and handling areas serve as the basis for administrative procedures or HERO emission control procedures to preclude HERO.

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Frequency Ranges (MHz)	SSD Equations
$0.01 \le f < 2.0$	SSD $= 1.37f\sqrt{P_t G_t}$ meters SSD $= 4.5f\sqrt{P_t G_t}$ feet
$2.0 \le f < 80.0$	SSD $= 2.74\sqrt{P_t G_t}$ meters SSD $= 9\sqrt{P_t G_t}$ feet
$80.0 \le f < 100{,}000$	SSD $= \left(\frac{219}{f}\right)\sqrt{P_t G_t}$ meters SSD $= \left(\frac{718}{f}\right)\sqrt{P_t G_t}$ feet

where:

SSD is the distance in the units designated.

P_t is the average power output of the transmitter in watts.

G_t is the numerical (far-field) gain ratio (not the dB value) of the transmitting antenna, derived as follows:

$$G_t = 1 \times 10^{G/10}$$

where:

G is the gain in dBi

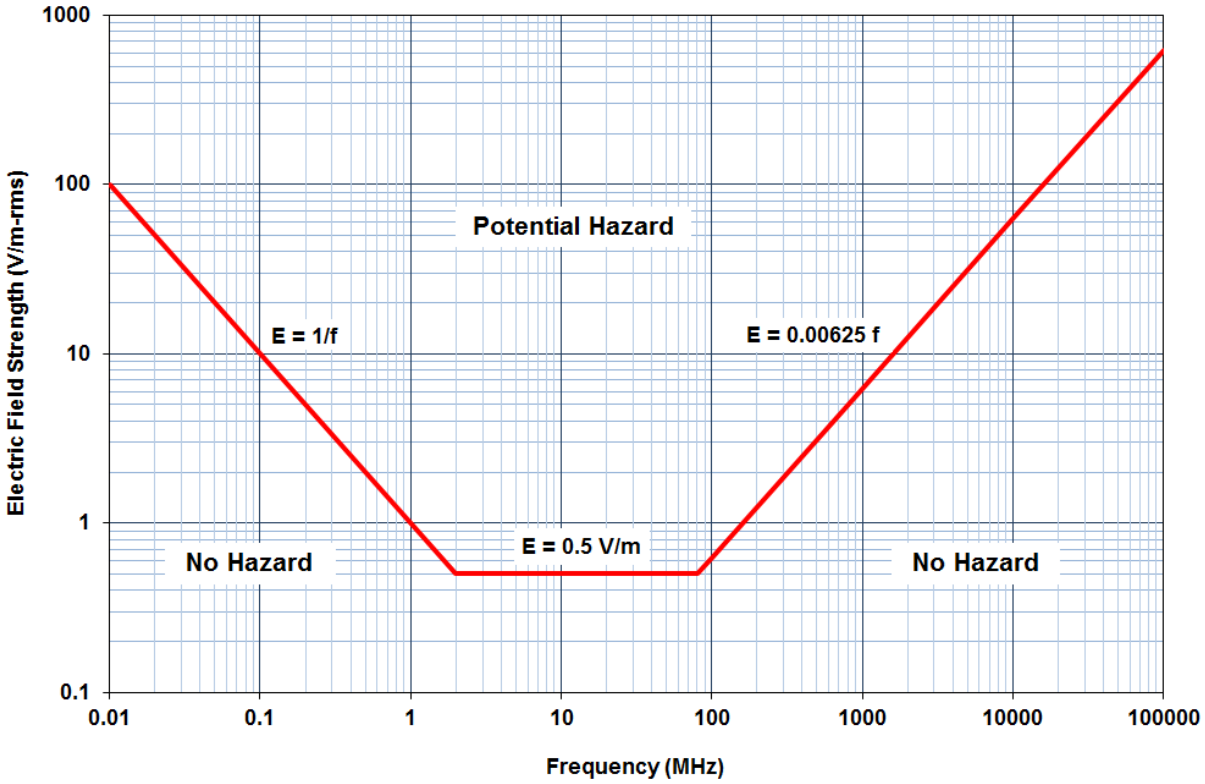
f is the transmitting frequency in megahertz (MHz)

Notes:

1. The information above represents “worst-case” conditions for safe distance required.
2. Equations are provided with the proper numerical multipliers to yield distances in either meters or feet.
3. Maintain a minimum SSD of 3 meters (10 Feet) when the computed SSD is less than 3 meters (10 feet); see 5.1.3, TABLE I.

FIGURE 1. Graph and equations for computing safe field strength/distance for HERO SUSCEPTIBLE ORDNANCE

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Frequency Ranges (MHz)	SSD Equations
$0.01 \le f < 2.0$	SSD = $5.5f\sqrt{P_tG_t}$ meters SSD = $18f\sqrt{P_tG_t}$ feet
$2.0 \le f < 80.0$	SSD = $10.95\sqrt{P_tG_t}$ meters SSD = $36\sqrt{P_tG_t}$ feet
$80.0 \le f < 100{,}000$	SSD = $\frac{876}{f}\sqrt{P_tG_t}$ meters SSD = $\frac{2{,}873}{f}\sqrt{P_tG_t}$ feet

where:

SSD is the distance in the units designated.

P_t is the average power output of the transmitter in watts.

G_t is the numerical (far-field) gain ratio (not the dB value) of the transmitting antenna, derived as follows:

$$\mathbf{G_t = 1 \times 10^{G/10}}$$

where:

G is the gain in dBi

f is the transmitting frequency in megahertz (MHz)

Notes:

1. The information above represents "worst-case" conditions for safe distance required.
2. Equations are provided with the proper numerical multipliers to yield distances in either meters or feet.
3. Maintain a minimum SSD of 3 meters (10 Feet) when the computed SSD is less than 3 meters (10 feet); see 5.1.3, TABLE I.

FIGURE 2. Graph and equations for computing safe field strength/distance for HERO UNSAFE ORDNANCE

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4.3 HERO Classifications. The HERO classifications are extremely important in that they immediately identify the level of HERO concern for the warfighter. For example, HERO SAFE ordnance, when encountered, implies that there are no RF concerns as long as the general HERO requirements for ordnance outlined are followed (see 7). HERO SUSCEPTIBLE and HERO UNSAFE ORDNANCE, when encountered, identify the need to manage HERO, and can be accomplished through HERO emission control procedures with specific controls to mitigate HERO during specific operations or through the use of SSDs (or a combination of both).

4.4 SSDs. HERO emission control procedures based on SSDs are more prevalent at ground/shore facilities and are used to manage HERO for aircraft, as well as portable and mobile emitters that have no fixed location. As described in 4.2, the distances can be calculated from transmitter/antenna systems characteristics (frequency, power, antenna gain of the emitter) and the HERO classification of the ordnance: HERO SAFE, SUSCEPTIBLE or UNSAFE. Aboard ship, the limited real estate generally does not allow precluding HERO by maintaining a HERO standoff distance and is not a practical solution. Consequently, measured data is used to better characterize the shipboard EME and to identify which transmitter/antenna systems will be secured when HERO UNSAFE or HERO SUSCEPTIBLE ordnance is present topside. The administrative work-around is implemented using HERO emission control procedures. As this is a more comprehensive approach, information is provided (see 6) on developing operational guidance and emission control procedures, determining HERO classifications, and the process for developing an operational guidance or HERO emission control procedures document.

In addition to using SSDs, shore facilities, on the other hand, are typically divided into HERO zones (7.3.2) such that when ordnance is present in a particular zone, the necessary EMCON conditions can be set. In this case, SSDs can be used to manage HERO within each of the HERO zones; although, measured data is generally used to help minimize the restrictions placed on transmitting systems since SSDs do not consider the effect of the terrain on the EME.

5. MANAGING HERO THROUGH SSDS

5.1 Introduction. In order to determine the calculated HERO SSD for an emitter, specific characteristics, such as average transmitter power, minimum operational frequency, and antenna gain must be identified. The MAE, or the limits provided in the HERO curves for ordnance classified as either HERO SUSCEPTIBLE or UNSAFE ORDNANCE, can be determined using the graphs in FIGURE 1 and FIGURE 2. As will be discussed, using the MAE based on the HERO classification provides a more conservative approach for managing HERO. Once the transmitter parameters are known and the HERO classification has been defined, FIGURE 1 and FIGURE 2 are used to determine the distance at which the EME should not create a hazard to the ordnance. After all the emitters have been evaluated, the SSDs are used to ensure that all ordnance items are not transported, stored or used within these distances. Note that providing HERO SSDs based on a HERO classification and a calculated SSD is purely analytical and is generally a more conservative approach for managing HERO and the result has generally more restrictions and provides less operational flexibility.

5.1.1 Calculating SSDs. FIGURE 1 and FIGURE 2 (Joint HERO Curves) have been prepared on the basis of worst-case conditions for all HERO SUSCEPTIBLE and HERO UNSAFE ORDNANCE; these curves may be used in determining the maximum RF environment for any ordnance system when specific MAE susceptibility data is not available or

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for ordnance that has not been HERO evaluated. See Appendix A for a description of how the Joint HERO curves were derived. However, specific MAEs can be used to calculate SSDs by applying the specific frequency-dependent MAE to the Friis free-space equation and solving for distance. Specific test data generally provide MAEs that are less restrictive than the MAEs presented in the general HERO curves shown in FIGURE 1 and FIGURE 2. The purpose of these curves is to provide a standard set of MAEs for HERO SUSCEPTIBLE and HERO UNSAFE ORDNANCE when no specific MAE data is available.

5.1.2 SSD Based on Ordnance MAE. Specific test data on individual items may indicate greater MAEs at some frequencies and consequently, support shorter SSDs than those shown on FIGURE 1. Equations (5-1) and (5-2) can be used to calculate SSDs when specific MAEs are available from test data and may be less restrictive (closer distances) than those based on the HERO SUSCEPTIBLE curve.

$$SSD = 5.48 \frac{\sqrt{P_t G_t}}{\text{MAE}} \text{ meters} \quad (5\text{-}1)$$

$$SSD = 17.97 \frac{\sqrt{P_t G_t}}{\text{MAE}} \text{ feet} \quad (5\text{-}2)$$

where:

P_t = the average power output of the transmitter in watts (W)
G_t = the numerical (far-field) gain (not the decibel (dB) value) of the transmitting antenna
MAE = maximum allowable environment in volts per meter (V/m)

5.1.3 Exceptions to the SSD Calculation Methodology. Traditionally, the calculation methodology found in FIGURE 1, FIGURE 2, or 5.1.2 provided results that did not allow for distances of less than 10 feet (referred to as the "10-foot rule"). This approach was implemented due to uncertainties in the near-field and reactive-field of an antenna and was consistent with HERO ground plane testing of ordnance (see MIL-HDBK-240-2). However, there are a number of exceptions whereby sources of RF emissions (some of which are unintended and some of which are low-power devices) are expected to be, or are required to be, closer than 10 feet to ordnance or used in storage, assembly, and build-up areas (for example, Automatic Identification Technology (AIT) devices such as wireless laptops, passive radio-frequency identification (RFID) and active RFID). These devices are generally very low output devices (that is, less than 1 W) and their proximity to ordnance and low-output power require different techniques for mitigating HERO. In fact, with very low output devices the result is often the relaxation of the 10-foot rule. TABLE I provides exceptions to the minimum safe separation distance requirement of 10 feet (3 meters) and is particularly useful for handheld devices radiating at less than 1 W in and around areas that have HERO UNSAFE or HERO SUSCEPTIBLE ORDNANCE.

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TABLE I. Minimum SSD Exceptions

MINIMUM SEPARATION DISTANCE (FT.)	HERO CLASSIFICATION		
	SAFE	SUSCEPTIBLE	UNSAFE
> 10	General HERO Requirements	Use Calculated Distance	Use Calculated Distance
5	0.5 < EIRP ≤ 5 W All Frequencies	EIRP ≤ 0.5 W Frequencies ≥ 100 MHz	0.025 < EIRP ≤ 0.1 W 200 MHz ≤ Freq < 1 GHz
1	0.1 < EIRP ≤ 0.5 W All Frequencies	0.025 < EIRP ≤ 0.1 W Frequencies ≥ 200 MHz	0.025 < EIRP ≤ 0.1 W Frequencies ≥ 1 GHz
0	EIRP ≤ 0.1 W All Frequencies	EIRP ≤ 0.025 W All Frequencies	EIRP ≤ 0.025 W Frequencies ≥ 100 MHz
HERO SAFE ORDNANCE			
For shipboard applications, maintain 1.5 meters (5 feet) from the vertical projection of a lowered deck-edge antenna with the transmitter operating at an average Effective Isotropic Radiated Power (EIRP) of 1000 W or less, provided all loading procedures have been completed. Maintain a minimum separation of 15 meters (50 feet) from any transmitting antenna during vertical replenishment (VERTREP) operations.			
$$EIRP = P_t \times G_t$$ where: EIRP is the effective isotropic radiated power in W. $\mathbf{P_t}$ is the average power output of the transmitter in W. $\mathbf{G_t}$ is the numerical (far-field) gain ratio (not the dB value) of the transmitting antenna, derived as follows: $$G_t = 1 \times 10^{G/10}$$ where: G = gain in dBi Example: If the antenna far-field gain is 2.1 dBi, the far-field gain ratio is $1 \times 10^{2.1/10} = 1 \times 10^{0.21} = 1.62$			

5.1.4 Managing the EME. If the methodology (use of SSDs to mitigate HERO) does not provide usable results due to limited real estate or provides over-restrictive results, then other techniques can be employed that provide more flexibility for managing the EME at ordnance locations. Measured data can be used to better characterize the EME, actual MAEs can be used to identify the limits for each ordnance item, and HERO emission control procedures can be used, in lieu of SSDs, to identify specific transmitter/antenna systems to be secured (or reduced in power) when HERO UNSAFE or HERO SUSCEPTIBLE ORDNANCE is present at specific locations. These administrative work-arounds are implemented using HERO emission control procedures and can include defining specific safe areas for ordnance operations, frequency and/or power management, and sectoring of RF emissions. As described in MIL-HDBK-240-6, HERO assessments are a means for achieving this.

5.1.5 Multiple Superimposed Field Calculations. Where there are multiple emitters transmitting in the frequency range of 2 to 80 megahertz (MHz), a method has been determined to calculate and reduce the combined field to an acceptable safe level. To determine if

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superimposed fields (no one of which, acting independently, is sufficient to create a hazard) present a combined hazard to the ordnance system, use the formula:

$$E_T = \sqrt{{E_1}^2 + {E_2}^2 + \cdots + {E_n}^2} \qquad (5\text{-}3)$$

where:

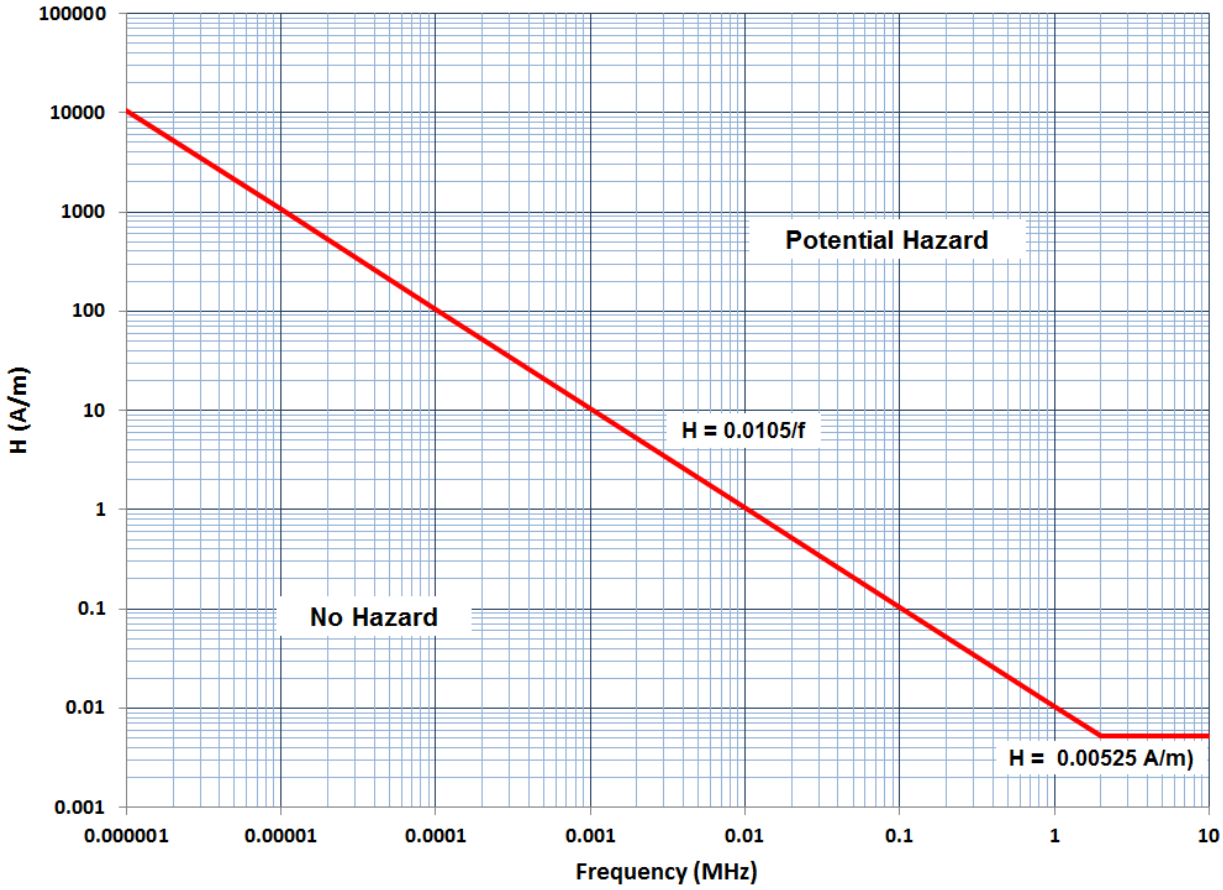
E_T is the total electric field (E-filed) strength to be calculated, and (E_1), (E_2), and so forth, are the individual field strengths to be added together.

NOTE:
The calculated total field should then be compared with the maximum allowable field. If it is less, no hazard exists.

If a hazard exists, at least one transmitter must be silenced or reduced in power and the preceding formula recalculated to determine if a safe condition now exists. If not, this process must be repeated until a combination of fields is found for which the total field is less than the maximum allowable field.

5.2 SSD Based on Magnetic Field (H-Field) Limits. Concern for adverse effects from H-fields was heightened with the introduction of the Electromagnetic Aircraft Launch System (EMALS) on a U.S. Navy aircraft carrier. Initial modeling and subsequent measurements of an operating EMALS confirmed the HERO impact to ordnance exposed to these unprecedented H-field levels. FIGURE 3 and FIGURE 4 are based on conservative assumptions for coupling of H-fields and apply to HERO SUSCEPTIBLE and HERO UNSAFE ORDNANCE, respectively. These figures are consistent with FIGURE 1 and FIGURE 2, in that the limits are presented in graphical format and tables of the associated SSDs. The figures may be used in determining the maximum allowable H-field environment for ordnance in proximity to systems characterized by low-frequency/high-current operation.

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Frequency Ranges (MHz)	SSD Equations
$0.000001 \le f < 2.0$	$\text{SSD} = \frac{\sqrt[3]{I*f}}{10640}$ meters $\text{SSD} = \frac{\sqrt[3]{I*f}}{34318}$ feet
$2.0 \le f < 10.0$	SSD = 57.3 meters SSD = 188 feet

where:

SSD = distance in the units designated

I = maximum current in nearby cable in amps

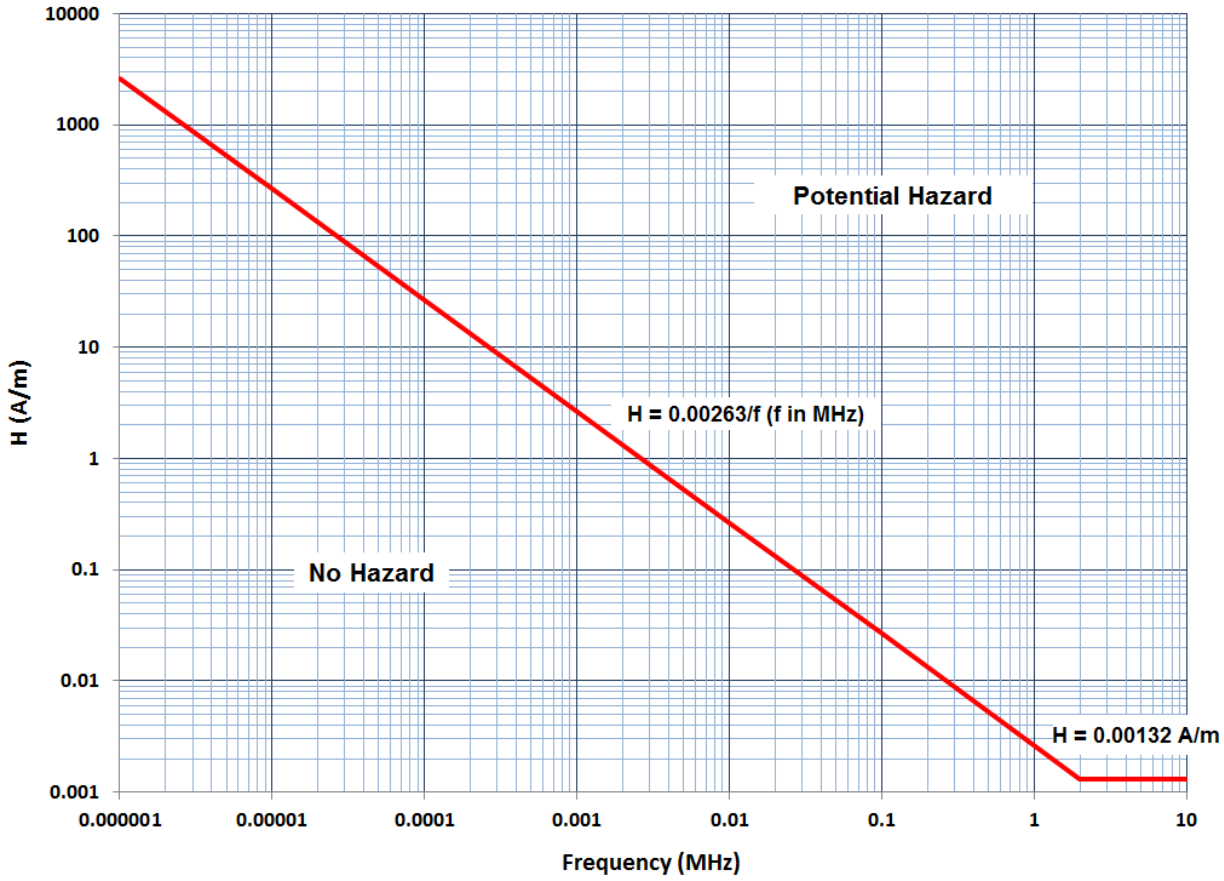
f = frequency in MHz

Notes:

1. The information above represents “worst-case” conditions for safe distance required.
2. Equations are provided with the proper numerical multipliers to yield distances in either meters or feet.

FIGURE 3. Graph and equations for computing safe field strength/distance for HERO SUSCEPTIBLE ORDNANCE

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Frequency Ranges (MHz)	SSD Equations
$0.000001 \le f < 2.0$	$\text{SSD} = \frac{\sqrt[3]{I*f}}{2660}$ meters $\text{SSD} = \frac{\sqrt[3]{I*f}}{8727}$ feet
$2.0 \le f < 10.0$	SSD = 90.9 meters SSD = 298 feet

where:

SSD is the distance in the units designated.
I = maximum current in nearby cable in amps.
f = frequency in MHz.

Notes:
1. The information above represents "worst-case" conditions for safe distance required.
2. Equations are provided with the proper numerical multipliers to yield distances in either meters or feet.

FIGURE 4. Graph and equations for computing safe field strength/distance for HERO UNSAFE ORDNANCE

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5.3 Risk Mitigation Techniques. Managing the HERO risk from E-fields is accomplished by maintaining E-field SSDs, limiting the magnitude of the E-field by reducing the output power of the emitter, or a combination of both. By using both the ordnance's HERO classification and the measured or calculated RF environment from emitter systems, determination of the best method can be made. In contrast, managing the HERO risk from H-fields is accomplished by maintaining H-field SSDs. Limiting the magnitude of the H-field by reducing system cable current levels may compromise the operation of the system and would therefore not be a viable option. Closed loop EID circuits allow current flow and may be susceptible to adverse effects from low frequency H-fields.

6. DEVELOPMENT OF HERO OPERATIONAL GUIDANCE AND EMISSION CONTROL PROCEDURES

6.1 Introduction. HERO emission control procedures are procedures for mitigating HERO that may include the use of procedural work-arounds, frequency and/or power management, sectoring, or managing specific operations and their locations which may include the use of SSDs. The purpose of HERO emission control procedures is to prescribe, through advance planning, the easiest and most efficient method of managing the conflict between the EME created by transmitting equipment and HERO-classified ordnance. In order to develop HERO emission control procedures, a number of actions and information is required:

a. Identification of ordnance present, their HERO classification, and associated MAEs.

b. Identification of transmitting systems, their locations with respect to ordnance, and the associated EMEs (measured and/or calculated).

c. Understanding of mission requirements.

In the end, the EMEs and SSDs associated with the transmitting systems are compared against the ordnance MAEs and their locations to determine the need for managing HERO. Once these have been determined, HERO emission control procedures can be developed to ensure that the operational EME is below the MAEs for ordnance.

6.2 Ordnance HERO Classifications. Ordnance containing EIDs are classified as either HERO SAFE, HERO SUSCEPTIBLE, or HERO UNSAFE ORDNANCE. These general HERO classifications provide a means for identifying a general level of susceptibility to the EME and are defined in MIL-STD-464 and Part 1 of this document.

6.2.1 How to Determine HERO Classifications. To determine the HERO classification of a particular ordnance item, refer to the Joint Ordnance or Service database which lists, by Department of Defense Identification Code (DoDIC) or Navy Ammunition Logistic Code (NALC), and/or National Stock Number (NSN), all ordnance items that have been evaluated for HERO. These databases also list the HERO classification (HERO SAFE, HERO SUSCEPTIBLE, or HERO UNSAFE) of each ordnance item containing EIDs. For items that have not been tested, see MIL-HDBK-240-2 for analysis techniques. The MAE tool can be used to determine the susceptibility to the EME; the absence of a HERO evaluation requires ordnance to be treated as HERO UNSAFE. FIGURE 5 is a flowchart to aid in determining HERO classification for an ordnance item.

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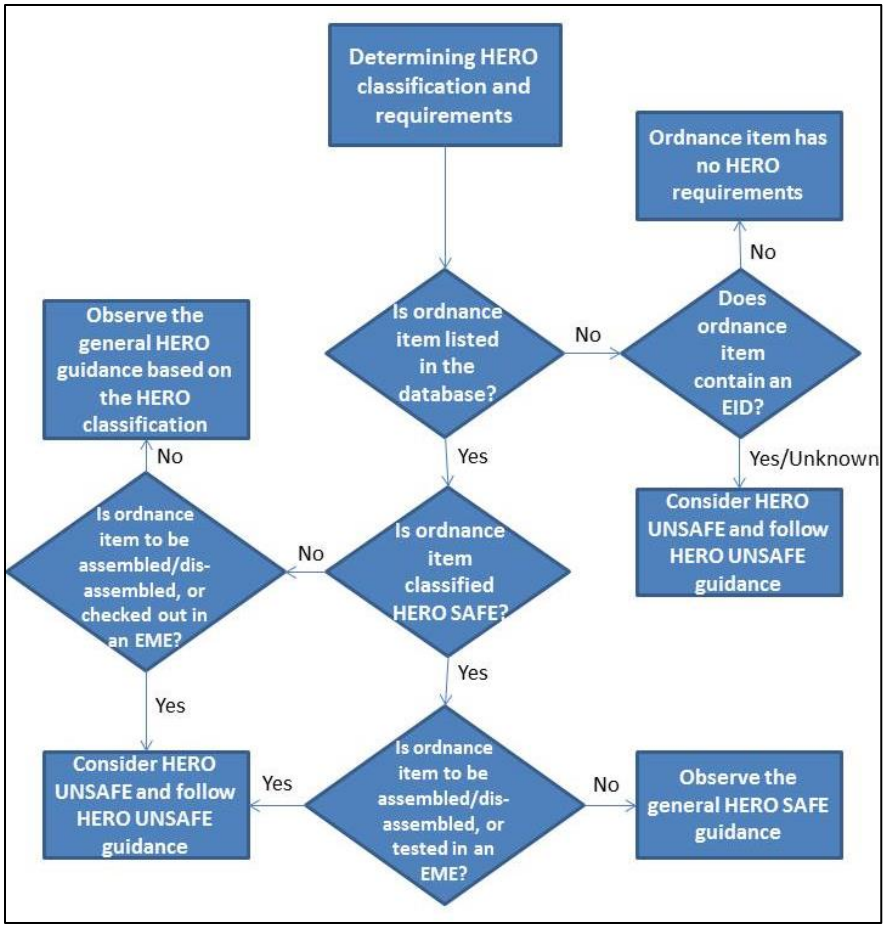


FIGURE 5. HERO classification guidance

If the ordnance item is listed as HERO SAFE, then only the general HERO guidance (see 7) must be followed. Further, if the ordnance item is listed as HERO SUSCEPTIBLE or HERO UNSAFE, then restrictions and/or risk mitigation procedures will be required. The datasheets stored in the ordnance database provide specific information about the HERO susceptibility of the ordnance item, including frequencies where the item is susceptible, MAE, EID firing consequence (safety/reliability), and the stockpile-to-safe-separation sequence (S4) phases (handling/loading, platform loaded, and so forth) where the item is susceptible to EMEs. In some instances, the data sheet also provides procedures to mitigate potential HERO problems. The Sample HERO Summary Datasheet is provided in Table VII, Part 2 of this document.

6.2.2 Conditions Leading to HERO UNSAFE Ordnance. Ordnance items normally classified as HERO SAFE or HERO SUSCEPTIBLE must be treated as HERO UNSAFE during assembly, disassembly, or testing, or if the items are subjected to unauthorized conditions and operations. Care should be taken to ensure that such conditions and operations occur in an area where the use of intentional RF emitters are restricted/controlled, and the ambient EME is well below the HERO UNSAFE curve presented FIGURE 2. Examples of conditions for HERO SAFE or SUSCEPTIBLE ORDNANCE that might lead them to become HERO UNSAFE are as follows:

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a. Assembly or disassembly of ordnance systems such as those undergoing modification, repair, upkeep, parts exchange, strike down, and so forth.

b. Tests involving additional electrical connections to the ordnance system such as resistance checks, continuity checks, and so forth.

c. Squibs, primers, blasting caps, impulse cartridges, and other EIDs having exposed leads or primer buttons that are unshielded and/or unfiltered, such as flash signals, igniters, tracking flares, and so forth, and have not been HERO-evaluated in these configurations.

d. Unshielded ordnance subassemblies, such as rocket motors, warheads, exercise heads, fuzes.

e. Damaged ordnance items that have internal components exposed, seams or joints that are no longer intact, or HERO shielding (such as barrier plates on 2.75-inch folding fin aircraft rocket (FFAR) launchers) breeched.

If the above conditions are encountered and exposure to EMEs cannot be avoided, HERO UNSAFE ORDNANCE can be protected by completely enclosing the item in a sealed, all-metal container.

NOTE: Ordnance items packaged in a wooden, cardboard, or plastic container or stacked on a metal pallet are not protected from the EME and must be treated as HERO UNSAFE ORDNANCE unless *their HERO classification states otherwise*.

Further, HERO UNSAFE ORDNANCE should only be removed from protective packaging in an EME that does not exceed the levels provided in FIGURE 2. Such areas may be found below decks of ships or in buildings designed with electromagnetic (EM) shielding. In areas where the EME does exceed the levels of the FIGURE 2, HERO UNSAFE ORDNANCE should not be permitted outside of its protective packaging unless appropriate emission controls are invoked.

Unexploded ordnance (UXO) is considered to be HERO UNSAFE ORDNANCE. The detection equipment used to locate UXO (that is, ground-penetrating radar, ground conductivity meters, and so forth) may be capable of generating sufficient EM energy to cause inadvertent actuation of EIDs in UXO. The Service technical point of contact (POC) listed in 10.5 can assist in the determination of the transmitting characteristics of the detection equipment.

6.2.3 Components of All-up-rounds (AURs). AURs that contain EIDs must be considered for HERO. Typically, HERO tests do not evaluate the individual components separate from the AUR unless operational requirements dictate the need (for example, blasting caps, ejector cartridges, and some bomb fuzes). Individual components that contain EIDs and are not listed in ordnance database are considered HERO UNSAFE ORDNANCE and require emission control procedures during all ordnance operations involving these components. This general guidance may be tailored if specific information is known about the individual component configuration and its projected use. The following provide additional guidance for tailoring the emission control requirements for individual components:

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a. Presence of Components/Subassemblies (transportation/storage phase). Components/subassemblies shipped in sealed, all-metal containers or metal foil packages are subject to only the general HERO requirements for HERO SAFE ORDNANCE. The unopened container/packaging may be safely/reliably transported, handled, or stored in the typical EME. See the appropriate technical manuals/orders for specific packaging or description of material composition of containers used for transporting and storing the item.

NOTE: The container/packaging must not be opened during this phase. Components/subassemblies that are shipped in non-metal containers or packages should be treated in accordance with their HERO classifications; if this is unknown, they must be treated as HERO UNSAFE ORDNANCE.

b. Removal of individual components/subassemblies from their containers, or packaging and assembly/disassembly of the components must be performed in an area designated for HERO UNSAFE ORDNANCE; otherwise, HERO emission control procedures are required.

c. Handling, removing, or installing components in subassemblies of AURs or assembly of AURs require HERO emission control procedures.

d. Once all the components have been assembled or installed into the system, HERO guidance is based on the HERO classification of the resultant AUR. Refer to the ordnance database for HERO classification of the AUR.

7. DEVELOPMENT OF HERO OPERATIONAL GUIDANCE DOCUMENT

7.1 Development Process. The HERO operational guidance document development process is shown in FIGURE 6. The three critical elements necessary for the development of the HERO operational guidance document are the emitter list, ordnance inventory list, and drawings highlighting ordnance storage, operations, and transportation locations. FIGURE 6 shows how these critical elements cumulate into a HERO operational guidance document. The following paragraphs describe each block of the flowchart and how it relates to the HERO operational guidance document development process.

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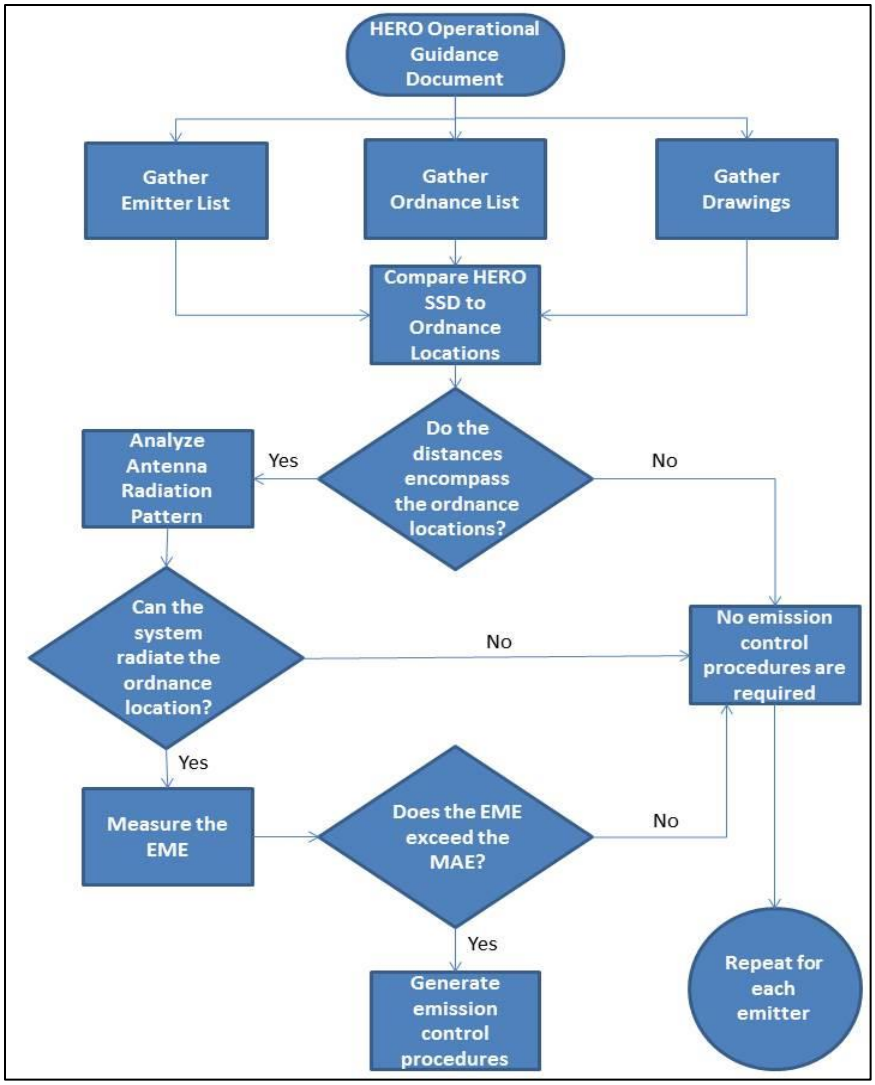


FIGURE 6. HERO operational guidance development process

7.1.1 Emitter List. The emitter list contains an inventory of all fixed, mobile, and portable emitter systems installed or used on the platform or facility. The two-fold importance of this listing is that it provides the user with the location designation for each transmitting antenna as well as the calculated HERO SUSCEPTIBLE and HERO UNSAFE ORDNANCE SSDs. These calculated distances are considered to be worst-case SSDs in that they assume main-beam illumination and maximum efficiency of the transmitter/antenna system as it relates to the RF output power (that is, no system losses); SSDs are derived for a category of ordnance (that is, SUSCEPTIBLE or UNSAFE) based on the worst-case susceptibility for that category and not on the specific susceptibility for an ordnance item. Most often, the HERO operational guidance document is developed using measurements of the EME and considers all documented transmitter systems and generally is not based on calculated SSDs. As such, the measurement-based HERO operational guidance document obtained from a HERO survey is more than a two-dimensional SSD paper study; it measures the true RF environment and considers system operation parameters (that is, elevation and azimuth restrictions), off-axis radiation levels, and their proximity to ordnance locations and the surrounding superstructure, buildings, or terrain.

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This data provides a less restrictive means for managing HERO and minimizes the impact on operations.

The important information included in the emitter list is as follows:

a. Antenna location or antenna number

b. Antenna nomenclature

c. Antenna type

d. Antenna gain (in decibels relative to isotropic (dBi))

e. Frequency range of the transmitter

f. Transmitter average power

g. Transmitter nomenclature

h. Calculated HERO UNSAFE distance

i. Calculated HERO SUSCEPTIBLE distance

The emitter listing should be updated whenever new emitter equipment is installed or existing emitter equipment is relocated or removed from service.

7.1.2 Ordnance List. The purpose of the ordnance list is to show each ordnance item's respective HERO classification (for example, No HERO Requirement (NHR), HERO SAFE, HERO SUSCEPTIBLE or HERO UNSAFE ORDNANCE). The items of HERO concern are those that contain EIDs, indicated by a HERO classification of HERO SAFE, HERO SUSCEPTIBLE or HERO UNSAFE ORDNANCE.

The ordnance inventory susceptibility listing is generated from the Services' or Joint ordnance databases. These databases also allow the user to access the ordnance data sheets. These data sheets provide a wealth of information on each item including overall HERO classification, MAEs for each applicable S4 phase, NSN, part numbers, packaging, and application on which the item has been tested or evaluated. It is important to note that an item's susceptibility can vary depending on its application and/or platform.

The ordnance list must be updated whenever new ordnance items are introduced or changes have occurred to existing ordnance items' HERO classifications. Ordnance items that are no longer used also should be annotated, and then it is simply a matter of removing them from the existing HERO operational guidance document.

7.1.3 Drawings. The purpose of the drawings is to show the locations of all the installed emitter equipment, as well as identify ordnance operation areas. Ordnance locations include all locations that are represented in the S4 phases; that is, transportation/storage, assembly/disassembly, staged, handling/loading, and platform-loaded.

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The drawings should be updated whenever there are changes to the fixed emitter configuration or when new ordnance operation areas are established or relocated. Emitter equipment that has been removed from service and ordnance operation areas that have been disestablished should also be annotated on the drawings.

7.2 Operational Guidance Development. The next step in the HERO operational guidance document development process is determining the emitter systems that require HERO controls. This is accomplished in a three-phased approach as described in the following paragraphs.

7.2.1 SSD Comparison. A comparison of the emitter's calculated HERO UNSAFE and HERO SUSCEPTIBLE ORDNANCE SSD with respect to ordnance operation areas is conducted to determine if the emitter equipment can potentially impact ordnance operation areas. These distances represent the worst-case condition for HERO UNSAFE and HERO SUSCEPTIBLE ORDNANCE. This comparison is accomplished by superimposing these SSD arcs onto the drawings to determine if the arcs impinge upon ordnance operation areas. The emitter equipment that does not impinge upon ordnance operation areas will not require HERO emission control procedures, and no further action is required. However, emitter equipment whose separation distance arcs do impinge upon ordnance operation areas is identified for further study and possible HERO emission control procedures. Further analysis is necessary because these calculations are worst-case and do not take into consideration antenna location, antenna orientation, antenna pattern, or system losses that can potentially reduce the efficiency of the system, thus reducing the EME levels produced. This, in turn, reduces the actual distances where the ordnance MAEs are exceeded.

7.2.2 Antenna Radiation Pattern Analysis. An analysis of the antenna's vertical location (that is, height above ground), vertical beam width, and azimuth radiation pattern, including any installed radiation cutouts in relation to ordnance operation areas, is conducted to determine if the emitter equipment can potentially impact ordnance operation areas. Emitter equipment that is determined to not be capable of illuminating ordnance operation areas does not require HERO emission control procedures. Systems that are capable of directly illuminating ordnance areas are identified for measurement in order to quantify the EME levels. The measured EME levels are used as the basis for formulating the appropriate HERO emission control procedures to manage the emitter equipment for HERO.

7.2.3 EME Measurements. EME measurements are taken during the HERO survey process (MIL-HDBK-240-6) and are used to formulate system-specific HERO emission control procedures. These HERO emission control procedures may end up being as restrictive as the SSD comparison, but usually result in less restrictive procedures.

7.3 HERO Emission Control Conditions. The next step in the process is to establish HERO conditions. HERO conditions are a set of HERO emission control procedures that are applicable to specific ordnance locations.

7.3.1 Platforms. For platforms, this could be a specific location or an area of the platform that requires a common set of HERO emission control procedures. Different locations may have identical HERO conditions as a result of this process; that is, the same emitter

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equipment will be restricted by HERO emission control procedures imposed in more than one location.

7.3.2 HERO Zones. When developing HERO emission control procedures for large areas that contain multiple ordnance locations, it is advantageous to assign a HERO zone identification to each of the ordnance areas. This will facilitate the setting of HERO emission control procedures. Creating HERO zones assists in managing the number of emitters that must be silenced during ordnance operations (S4 phases) and reduces the number of HERO conditions. Ordnance locations that are similar in type, geography and emission control restrictions are grouped, or zoned, into localized areas in order to simplify the application of HERO emission control procedures. Roadways and landmarks are used to designate the zone perimeter. Some typical examples of these zones are as follows:

a. Ordnance storage magazine areas.

b. Ordnance assembly/disassembly areas.

c. Runway and hangar areas, including Combat Aircraft Loading/Parking Area, arm/de-arm, hot cargo, and red label areas, and parking aprons.

d. Administrative areas through which ordnance is transported to/from runways, magazine areas, or gates.

7.4 HERO Emission Control Procedures Document. The HERO operational guidance document implements the HERO Program and is typically a two-part document. The first part consists of procedures for managing HERO and setting HERO emission control procedures during ordnance operations. The second part is an administrative section that assigns responsibilities for establishing/maintaining a HERO Program, monitoring the HERO posture of the platform or facility, and procedures for setting HERO emission control procedures.

7.4.1 Procedures for Managing HERO. This part of the document will consist of the summary of the information gathered from 7.3 and will provide specific guidance for each DODIC/NSN (or a general class of ordnance, such as all HERO SUSCEPTIBLE ORDNANCE) in a particular location.

7.4.2 Administrative Information. This part of the document will consist of information relating to the HERO Program being imposed for the platform/facility. It will consist of topics such as:

a. Referencing the policies and procedures with regard to HERO.

b. Providing a general description of the HERO program.

c. Designating responsibilities and points of contacts with respect to HERO and electromagnetic emissions.

d. General HERO Requirements.

e. General procedures for managing HERO.

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8. GUIDANCE FOR MANAGING HERO DURING NATO OPERATIONS

8.1 Introduction. This section covers the procedures for controlling HERO during Allied Operations or exercises involving U.S. and NATO forces. These procedures are not intended to replace the ordnance HERO classifications or HERO emission control procedures described elsewhere, or within the specific HERO survey reports, but are provided so that HERO can be appropriately addressed when operating in a NATO environment. Unlike the U.S., most NATO nations do not assign classifications (that is, SAFE, SUSCEPTIBLE, UNSAFE) to ordnance or manage HERO through HERO emission control procedures. Instead, Susceptibility Radiation Hazard [RADHAZ] Designator (SRAD) codes are assigned to ordnance, which define a level of susceptibility for ordnance. Transmitter RADHAZ Designator (TRAD) codes are assigned to emitters, which define a level of radiated emissions for emitters. Once assigned, the two are compared to identify HERO concerns and establish SSDs.

8.2 NATO RADHAZ. NATO Radio and Radar Radiation Hazards (RADHAZ) procedures are fully defined in the Allied Environmental Conditions Publication (AECP)-02/Multinational Environmental Conditions Publication (MECP)-02, "NATO NAVAL RADIO AND RADAR RADIATION HAZARDS MANUAL," and Supplement. The U.S. accepted use of these procedures by ratification of Standard NATO Agreement (STANAG) 1380. As previously mentioned, there are some significant differences in the methods of AECP-02/MECP-02 and the current U.S. approach to defining and implementing HERO emission control procedures. It is the intent of this section to introduce the NATO RADHAZ methodology, explain the differences between the NATO and U.S. approach to HERO mitigation, and to provide a means for integrating the two approaches. The basis for determining and setting HERO emission control procedures in a NATO environment is the use of SRAD and TRAD codes as described in the following paragraphs.

8.3 Ordnance SRAD Codes. AECP-02/MECP-02 does not use the HERO classifications of HERO SAFE, HERO SUSCEPTIBLE, and HERO UNSAFE ORDNANCE. Rather ordnance susceptibilities are given a 14-character alphanumeric code, in seven two-character sets, called an SRAD code that identifies the general susceptibility of an ordnance item or a platform containing ordnance. The first character of each set of the SRAD code is an upper-case letter that identifies the frequency range of the susceptibility. The letters used are "R," "T," "U," "V," "WA," "WB," "YA," "YB," and "Z," covering frequencies from 150 kHz to 45 gigahertz (GHz). The second character of each set of the SRAD codes is a number that describes the level of susceptibility in that frequency range. The indices are from zero, being the most susceptible to RF energy, to seven being the least susceptible. An index of zero would compare to a HERO classification of HERO UNSAFE ORDNANCE. An index of seven would compare to a HERO classification of HERO SAFE ORDNANCE. A complete SRAD code for an ordnance item will contain nine alphanumeric sets, such as R3 T1 U5 V0 WA2 WB3 YA6 YB5 Z4. In the event all indices are the same (for example, R7 T7 U7 V7 WA7 WB7 YA7 YB7 Z7), the SRAD is simplified to read "SRAD = 7." AECP-02/MECP-02 shows the distribution of frequencies within the bands for each letter of the alphanumeric code and the MAE for each index number. To summarize, the letter identifies the frequency band and the index number identifies the MAE within each band to which the ordnance may be safely exposed.

As an update to AECP-02/MECP-02, TABLE II has been modified to allow further refinement of specific guidance (sub-bands) for the frequencies in the W and Y bands. The W band covers

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the frequency range from 790 MHz to 4.5 GHz and was broken into two sub-bands (WA) 790 MHz to 2.7 GHz and (WB) 2.7 to 4.5 GHz. The Y band covers the frequency range from 4.5 to 18 GHz and was also broken into two sub-bands (YA) 4.5 to 8.5 GHz and (YB) 8.5 to 18 GHz. The SRAD codes can be developed using either the entire W and Y bands (e.g., R3 T1 U5 V0 W2 Y6 Z4) or redefined to the sub-bands (for example, R3 T1 U5 V0 WA2 WB7 YA6 YB7 Z4). The benefit of splitting the W and Y bands into smaller bands is that a single emitter that operated in a narrow frequency range will not drive the code for the entire band. The use of W and Y sub-bands is optional. However, using the new sub-bands will reduce the emission control restrictions placed on emitter systems and/or ordnance items. As a result, it should minimize the impact to the U.S./NATO/Allied operations and improve mission success.

TABLE II. SRAD Code Schematic*

Susceptibility Index	Field Intensity (V/m)			Power Intensity (W/m^2)					
	R 150 kHz 600 kHz	**T 600 kHz 32 MHz**	**U 32 MHz 150 MHz**	**V 150 MHz 790 MHz**	**WA 790 MHz 2.7 GHz**	**WB 2.7 GHz 4.5 GHz**	**YA 4.5 GHz 8.5 GHz**	**YB 8.5 GHz 18 GHz**	**Z 18 GHz 45 GHz**
7	200	300	200	400	1500	10000	8000	10000	800
6	200		150	200	1250	4000	4000	4000	600
5	100			26	1000				500
4	60			10	400				
3	20			1	200				
2	10			0.26	100				
1	6	3		0.1	10		40		50
0	2	0.5		0.05	1		10		

** This table has been extracted from AECP-02/MECP-02 and reformatted for use in this document.*

As outlined in AECP-02/MECP-02, each nation is responsible for determining the SRAD codes for its ordnance and weapon systems containing EIDs. SRAD codes for ordnance are determined by the HERO Program authorities; these codes can be derived from the HERO classifications. The information is provided in the data sheets (available in the Joint database). Ordnance personnel, aircraft crews, or others who may be transferring ordnance from one NATO platform to another are responsible for providing SRAD codes to the host nation's platform. Aircraft in extreme circumstances, approaching another NATO platform, are expected to transmit their SRAD code at a distance of 5000 meters. The SRAD code for a platform, such as an aircraft, would be based upon the most susceptible ordnance item on that platform within each frequency range and is referred to as a "PLATFORM SRAD CODE."

8.4 Emitter TRAD Codes. Unlike the U.S. HERO Program, AECP-02/MECP-02 also provides a means for transmitter/antenna systems to be given classifications similar to ordnance. These codes are referred to as TRAD codes. Similar to an SRAD code, the alphanumeric TRAD code consists of a leading letter identifying the frequency range followed by an index number identifying the effective radiated power (ERP) of the system. An individual emitter system will usually have only one alphanumeric set, as opposed to seven sets for ordnance due to its generally narrow frequency band; although more than one alphanumeric set is possible, the

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TRAD code with the highest index should be used. The numerical index of the TRAD code is determined by the average power and the antenna gain using AECP-02/MECP-02.

8.5 AECP-02/MECP-02 RADHAZ Procedures. When using SRAD and TRAD codes, the basic procedure for RADHAZ control is through the comparison of the SRAD code of the approaching ordnance and/or platform to the TRAD code of the receiving or host platform. If, in any circumstance, the TRAD code exceeds the SRAD code in a particular frequency range, the potential exists that the emitter system's EME is greater than the MAE of the ordnance. If it is possible to relocate the ordnance or weapon system, AECP-02/MECP-02, Table G-1 may be used to determine a minimum safety distance (in meters) between the ordnance/weapon and the transmitting antenna. Refer to AECP-02/MECP-02 for guidance. It is important to note that the SSD between a transmitter/antenna and a specific ordnance item as identified in AECP-02/MECP-02 will be somewhat larger than the U.S. HERO Program approach to calculating a SSD for the equivalent transmitter/antenna system and an ordnance item with the same susceptibility. This is because the results in AECP-02/MECP-02 are the result of an ERP based on a range of values, not a calculation based on the transmitter/antenna system's specific characteristics.

Similar to the U.S. HERO approach, when SSDs are not a viable solution, AECP-02/MECP-02 prescribes other options such as the following:

a. Reducing power.

b. Sector blanking.

c. Finding an alternate landing position that provides the minimum safety distance.

d. If possible, jettisoning the susceptible ordnance.

8.6 U.S. Emission Control Procedures. The U.S. approach to HERO is to evaluate ordnance items and determine the MAEs through HERO testing, determine the EMEs through instrumented HERO surveys, and, based on the information obtained during HERO tests and surveys, provide specific HERO emission control procedures. HERO tests and evaluations determine an item's susceptibility or MAE and provide a HERO classification. Platforms and installations are "surveyed" for HERO to determine the actual EME at ordnance locations and transportation routes. If measured EMEs do not exceed the MAEs for specific ordnance items, then there are no HERO restrictions. If the EMEs exceed the MAEs for specific items, HERO emission control restrictions are required. These HERO emission control restrictions are provided in HERO emission control procedures and can take the form of maintaining SSDs; silencing the offending emitter; frequency or power management; sectoring; or implementing operational controls on either the weapon and/or the emitter. HERO emission control procedures provide HERO conditions that specifically identify all of the emitters that must be restricted for ordnance items of concern. The U.S. HERO process for determining HERO emission control restrictions goes one step further than the procedures of AECP-02/MECP-02 to ease the setting of HERO emission control procedures; in essence, making the process of calculating SRAD and TRAD codes transparent and going directly to a defined HERO emission control condition with specific HERO restrictions.

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9. DOCUMENTATION

9.1 Introduction. When it is necessary for contractors to obtain the data to show compliance with the Services' requirements, use 7.4 for the documentation requirements.

9.2 Purpose of Operational Guidance Documentation. Operational Guidance documentation incorporates the total assimilation of the program. Without proper documentation, the operational guidance may be unclear leading to an unwanted event (for example, dudding the weapon, detonating a warhead, firing a rocket).

10. NOTES

10.1 Intended Use. This handbook can be used by any facility that has a need to document the HERO procedures for any of the military departments.

10.2 Supersession Data. Not applicable.

10.3 Subject Term (Keyword) Listing.

Electromagnetic (EM)
Electromagnetic Environmental Effects (E3)
Electromagnetic Radiation Hazards
Emission Control
Hazards of Electromagnetic Radiation to Ordnance (HERO)
Ordnance
North Atlantic Treaty Organization (NATO)
Survey
Susceptibility Radiation Designator (SRAD)
Transmitter Radiation Designator (TRAD)

10.4 Identification of Changes. Not applicable.

10.5 Technical POCs. Requests for additional information or assistance on this standard can be obtained from the following:

Air Force

Air Force Safety Center (HQ AFSEC/SEW)
9700 G Avenue, SE
Kirtland AFB, NM 87116
(505) 846-0450
HQAFSCSEW@us.af.mil

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85th Engineering Installation Squadron (EIS/SCYM)
670 Maltby Hall Drive
Building 7701
Keesler AFB, MS 39534
(228) 377-3920
85eis.scym.1@us.af.mil

Army

US Army Test and Evaluation Command (HQ, ATEC)
6617 Aberdeen Blvd., Building 2202
ATTN: CSTE-OPS
Aberdeen Proving Ground, MD 21005-5001
(443) 861-9664
usarmy.apg.atec.mbx.atec-hq-g3-5@mail.mil

Army Test and Evaluation Command (ATEC)
E3 Test Division (E3 for Missile System Functions)
ATTN: TERT-ECE
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APPENDIX A
JOINT HAZARDS OF ELECTROMAGNETIC RADIATION TO ORDNANCE (HERO) CURVE CREATION

A.1. Introduction. The Services have traditionally used their specific standard maximum allowable environment (MAE) curves published in their respective publications. The Joint Ordnance Commander's Group (JOCG) HERO Subgroup recognized the need to standardize these curves into a single Joint agreed-upon chart. In order to develop a "worst-case" MAE curve for any given ordnance system that would satisfy the needs of all the Services, the JOCG HERO Subgroup needed to understand the existing methods used by each Service to develop MAEs and provide HERO guidance to their respective personnel. Each JOCG Service representative provided an explanation and reference of how they currently perform HERO assessments to determine specific MAEs and guidance for their respective Services. Figure A-1 was extracted from the Army, Navy, and Air Force documents during the 1990s and provides graphs of the "worst-case" MAEs used by each of the Services prior to this initiative. The graph plots the MAE in terms of electric field (E-field) strength (E) in volts root mean square per meter (Vrms/m) versus frequency in megahertz (MHz).

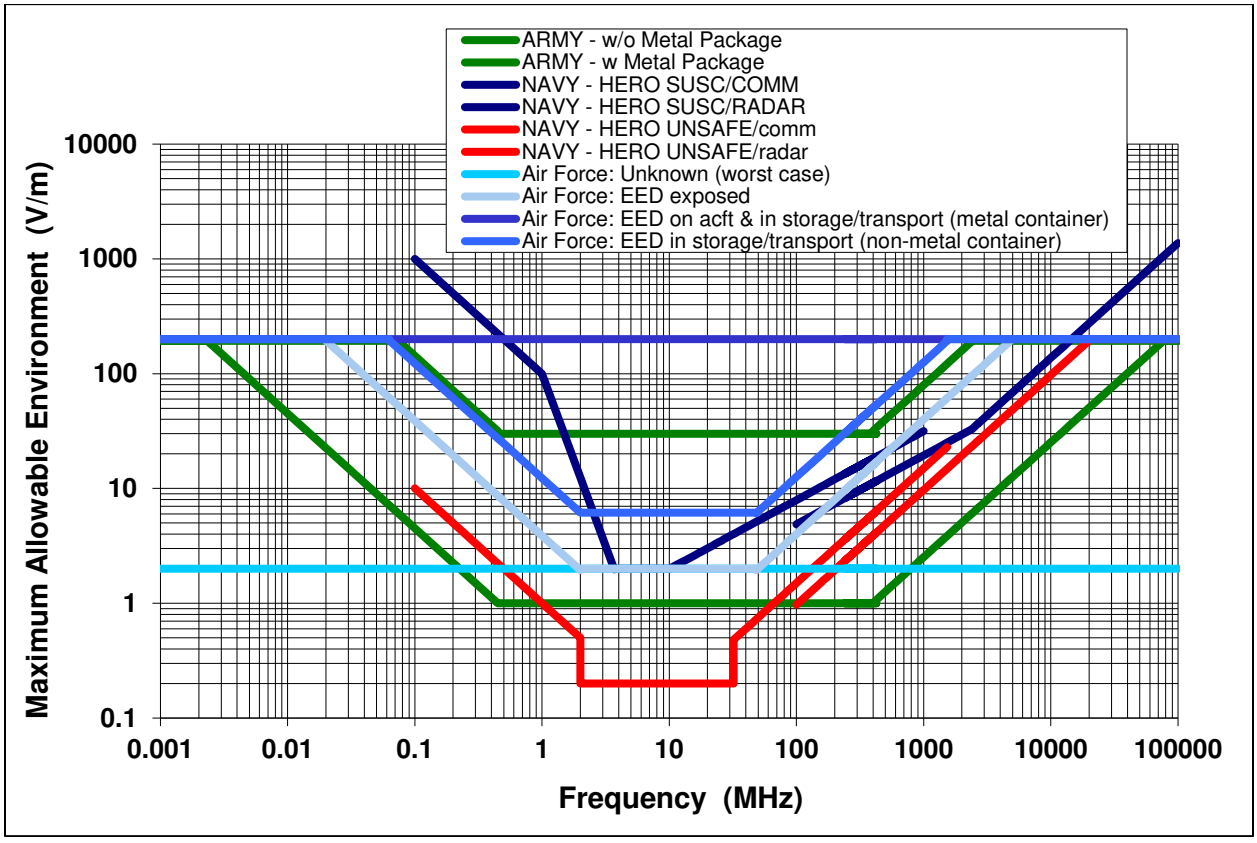


FIGURE A-1. Services' worst-case MAEs

As depicted in the graph, there is a wide range of MAEs recommended for a given ordnance system by each of the Services; however, the general shape of the curves is essentially the same. In addition, these graphs do not necessarily address all of the factors that affect the MAEs

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imposed on a given system, such as physical configuration, electrically initiated device (EID) sensitivity, firing consequence, or stockpile-to-safe-separation sequence (S4) phase. All of these parameters are important when determining specific MAEs for a particular system and should be considered when performing assessments and developing Service guidance.

Figure A-2 illustrates a "worst-case" MAE curve developed for use by all Services. This curve establishes the maximum EME levels to which an ordnance system can be exposed for which there is little or no known information about the item except that it contains an EID. This curve was both theoretically and empirically derived and assumes the following:

a. The ordnance system's design is such that the EID (including its firing circuit) is afforded little or no quantifiable electromagnetic interference (EMI) shielding protection.

b. There is an impedance match between the EID load and the antenna, thereby maximizing the energy transfer.

c. The EID is a very sensitive device (85-milliampere (mA) maximum no-fire current (MNFC) and 0.054 watts (W) maximum no-fire power (MNFP)).

d. The consequence of an inadvertent firing of the EID would result in a hazardous situation (safety).

NOTE: There are a few EIDs utilized within the Department of Defense (DoD) possessing firing sensitivities that are less than those stated above; however, they were not used in developing the "worst-case" MAE curve for several reasons. First, they constitute less than 1 percent of the total EIDs used within the U.S. military; second, they are usually very well-protected from the external electromagnetic environment (EME); and third, because of their sensitivity, they are very rarely (if ever) used in situations where their firing consequence would result in an immediate hazard but would result in performance degradation of the system.

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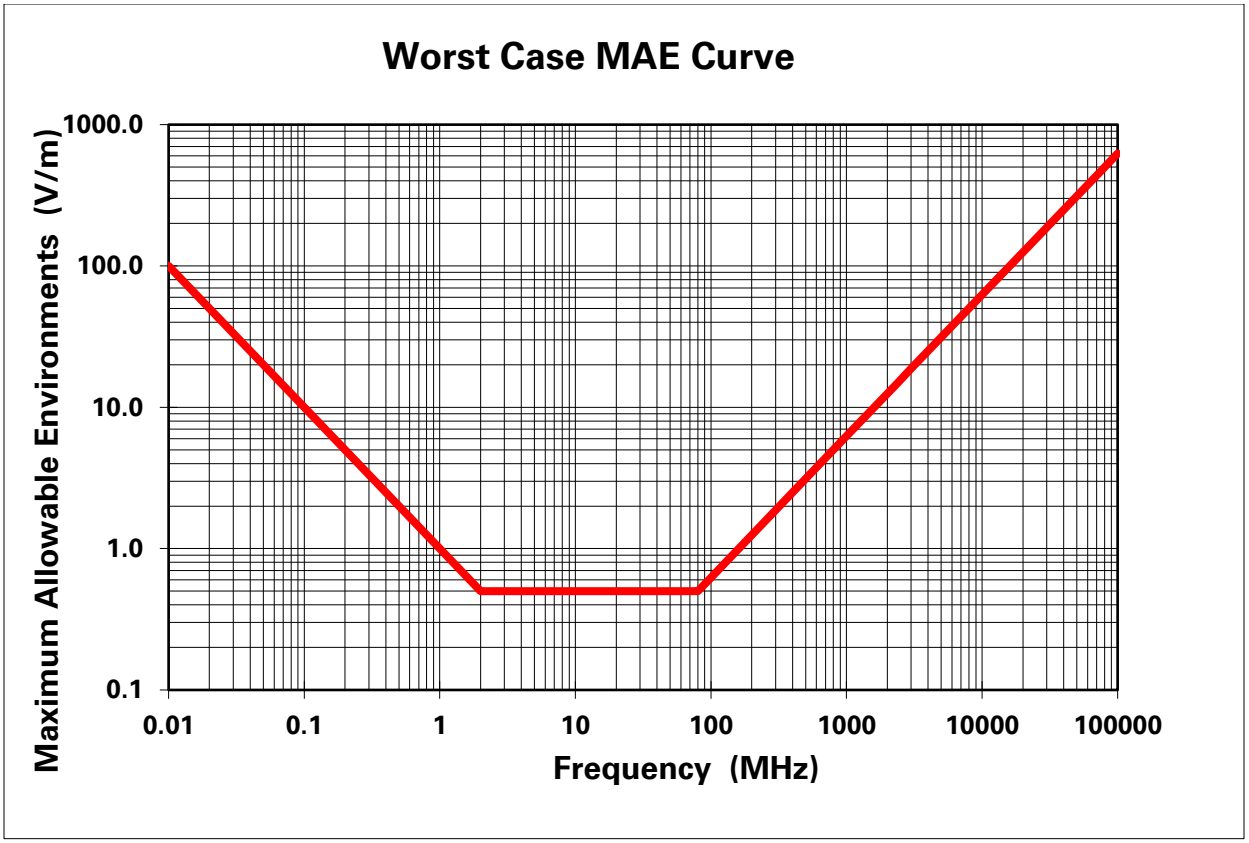


FIGURE A-2. Worst-case MAE curve

The following paragraphs provide discussions on the development of each segment of the curve and explanations of how they were derived.

A.2. Right-Hand Segment. The right-hand segment is theoretically based on a half-wave dipole antenna model. It assumes that for each frequency of concern the EID leads form a half-wave resonant dipole antenna (that is, each EID lead is one-quarter of the wavelength). It also assumes that the antenna/transmission line impedance is matched to the EID bridgewire circuit impedance and that the transmission line loss is negligible (that is, there is maximum transfer of energy from the antenna to the bridgewire). Figure A-3 depicts the half-wave resonant dipole model.

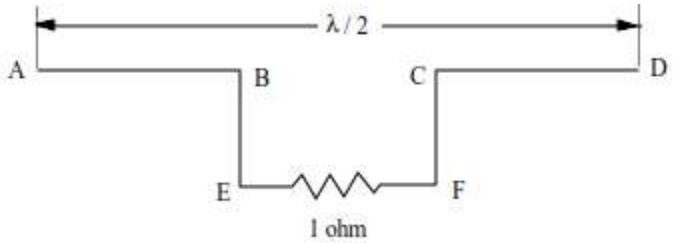


FIGURE A-3. Half-wave resonant dipole model

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The following equations are used to compute the "worst-case" MAE defined in terms of E-field strength (E):

$$E = \sqrt{PD_{MAE} * Z_0} \quad \text{(A-1)}$$

where:

E is the maximum E-field strength in volts root mean square per meter (Vrms/m)

PD_{MAE} is the maximum per density at the ordnance system/receive antenna in watts per square meter (W/m^2)

Z_0 is the intrinsic impedance of free space and is equal to 120π or approximately 377 ohms (Ω)

$$PD_{MAE} = \frac{P_{MNFP}}{A_e} * FCF \quad \text{(A-2)}$$

where:

P_{MNFP} is the maximum no fire of the EID in W

FCF is the firing consequence factor

A_e is the antenna effective area in square meters (m^2)

$$A_e = \frac{G_r * \lambda^2}{4 * \pi} \quad \text{(A-3)}$$

where:

G_r is the gain of the dipole antenna and is equal to 2.1 dBi or 1.64 gain ratio

λ is the wavelength (in meters) of the frequency defined in equation A-5

$$PD_{MAE} = \frac{P_{MNFP} * 4 * \pi * l}{\lambda^2 * G_r} * FCF \quad \text{(A-4)}$$

where:

P_{MNFP} is the maximum no fire of the EID in W

l is the system loss, which is negligible

λ is the wavelength (in meters) of the frequency defined in equation A-5
G_r is the gain of the dipole antenna and is equal to 2.1 dBi or 1.64 gain ratio

FCF is the firing consequence factor

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$$\lambda = \frac{c}{f} \quad \text{(A-5)}$$

where:

c is the speed of light in meters per second and is approximately 3 x 10^8

f is the frequency in MHz

Because the right-hand segment assumes the EID's electrical leads form a half-wave resonant dipole antenna, the wavelength (λ) is equal to two times the length of the antenna or four times the dimension of the ordnance item.

$$\lambda = 4 * d \quad \text{(A-6)}$$

where:

d is the length of one of the EID leads that forms part of a half-wave dipole antenna (in meters)

$$d = \frac{c}{4*f} \quad \text{(A-7)}$$

where:

c is the speed of light in meters per second and is approximately 3 x 10^8

f is the frequency in MHz

$$PD_{MAE} = \frac{P_{MNFP}*\pi*l}{4*d^2*G_r} * FCF \quad \text{(A-8)}$$

where:

P_{MNFP} is the maximum no fire of the EID in W

l is the system loss, which is negligible

d is the length of one of the EID leads that forms part of a half-wave dipole antenna (in meters)

G_r is the gain of the dipole antenna and is equal to 2.1 dBi or 1.64 gain ratio

FCF is the firing consequence factor

For the "worst-case" curve, the position of the right-hand segment on the graph is fixed; however, for a known ordnance system, it will vary as a function of the MNFP sensitivity of the EID, the firing consequence factor, and the system's shielding effectiveness (SE). The lower MAE limit of the right-hand segment is dictated by the middle segment and/or the physical dimension of the system.

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A.3. Middle Segment. The middle segment was derived empirically using data obtained by all three Services during the past 30 years of testing. An analysis of the data determined that for all systems tested, the maximum energy coupled to an EID occurred in the high-frequency (HF) range when testing a 2.75-inch rocket on a fixed-wing aircraft during simulated loading operations. The 2.75-inch rocket contained one EID with a 300-mA MNFC and a bridgewire resistance of 1.0 ± 0.1 omega (Ω). The rocket/EID uses a coaxial firing system design whereby the rocket case is the ground return path. When an aircraft is situated on a conductive surface (such as an aircraft carrier flight deck) in proximity to an active HF antenna, an extremely high energy potential can exist between the aircraft's outer skin and the deck. As a result, large amounts of electromagnetic radiation (EMR) current can be coupled into systems that utilize a coaxial firing design, causing the EID to fire. The middle segment was established using a transfer function derived from this HF model and the 2.75-inch test data.

For the "worst-case" curve, the MAE value of the middle segment is fixed at 0.5 Vrms/m from 2 to 80 MHz, which is based on the 85 mA MNFC. However, for a known ordnance system, the MAE value will vary depending on the MNFC sensitivity of the EID of concern, the firing consequence factor, and the system's SE.

A.4. Left-Hand Segment. The left-hand segment was derived both theoretically and empirically. As depicted on Figure A-2, the left-hand segment MAE increases as frequency decreases at a rate equivalent to 20 decibels (dB) per decade of frequency. The lower MAE limit of the left-hand segment is bounded by the value of the middle segment. For the "worst-case" curve, the position of the left-hand segment on the graph is fixed; however, for specific ordnance systems, it will vary depending on the resonant frequency as determined by four times the maximum physical length of the system.

A.5. MAE Data. Table A-1 reflects the MAEs for SUSCEPTIBLE and UNSAFE ordnance from the resulting work on the curves in the MIL-STD-464 Maximum External EME Levels for Ordnance table.

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TABLE A-1. Default MAEs

Frequency (MHz)		SUSCEPTIBLE MAE (V/m)*	UNSAFE MAE (V/m)*
0.01	2	2	0.5
2	30	2	0.5
30	150	2	0.5
150	225	3.8	0.9
225	400	5.6	1.4
400	700	10	2.5
700	790	18	4.4
790	1000	20	4.9
1000	2000	25	6.3
2000	2700	50	13
2700	3600	68	17
3600	4000	90	23
4000	5400	100	25
5400	5900	135	34
5900	6000	148	37
6000	7900	150	38
7900	8000	198	49
8000	8400	200	50
8400	8500	210	53
8500	11000	213	53
11000	14000	275	69
14000	18000	350	88
18000	50000	450	113

* V/m = volts per meter

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CONCLUDING MATERIAL

Custodians:
Army – SY
Navy – SH
Air Force – 11
DISA – DC5

Preparing Activity:
DISA (DC5)
(Project EMCS-2022-024)

Review Activities:
Army – AC, AM, AR, AT, AV, CE, CR, GL, MD, MI, PT, TE
Navy – AS, CG, EC, MC, OS, YD
Air Force – 13, 19, 22
DTRA – DS

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